

SOC Fundamentals

A complete, beginner-friendly guide to the Security Operations Centre - the SOC models, the analyst's toolkit, and every answer.

Course: SOC Fundamentals | Platform: LetsDefend | Free badge course

Level: Beginner | Topics: SOC models, roles, SIEM, log management, EDR, SOAR, threat intel

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About this walkthrough

This document walks through every lesson of the LetsDefend course SOC Fundamentals. It explains what a SOC is and how it is organised, what a SOC analyst does each day, and the tools every analyst relies on - SIEM, log management, EDR, SOAR and threat intelligence. Every lesson question and every quiz question is answered.

What you will learn

- The four SOC models and how a SOC team is structured.
- The roles inside a SOC, from analyst to SOC manager.
- The daily responsibilities and the skills a SOC analyst needs.
- How SIEM, log management, EDR, SOAR and threat intel fit together.
- The common mistakes analysts make - and how to avoid them.

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Before you start

SOC Fundamentals is a theory course - there is no machine to deploy. The practical answers come from the LetsDefend practice pages (Monitoring, Log Management, Endpoint Security and Threat Intel). Some lessons end with a simple checkpoint instead of a question; the ones that do ask are answered below. Try each question yourself before reading the answer.

LESSON 1

Introduction to SOC

The course opens by defining the thing everything else hangs off. A Security Operations Centre (SOC) is a facility where the information security team continuously monitors and analyses the security of an organisation.

Its purpose is to detect, analyse and respond to cyber security incidents by combining technology, skilled people and established processes. The three pillars work only together: detection without response is noise, and response without detection comes too late.

The three pillars of a SOC

Pillar	What it means in practice
Detection	Watch alerts, logs and feeds for signs of an attack.
Analysis	Investigate an alert to decide whether it is a real threat or a false positive.
Response	Contain the incident, eradicate the cause and recover affected systems.

Q1. Introduction to SOC - complete the lesson.

ANSWER

No answer needed - read the lesson and click Check.

LESSON 2

SOC Types and Roles

Not every organisation runs the same SOC. Depending on its security needs and budget it chooses one of four delivery models.

SOC model	How it works
In-house SOC	The organisation builds and runs its own cyber security team. It needs a budget that can keep the SOC running continuously.
Virtual SOC	No permanent facility - the team works remotely from different locations.
Co-Managed SOC	Internal SOC staff work together with an external Managed Security Service Provider (MSSP). Coordination is key.
Command SOC	Oversees several smaller SOC's across a large region - typical of big telecom providers and defence agencies.

People, process and technology

A successful SOC needs strong coordination between three things:

- People - highly trained staff who know security alerts and attack scenarios and can adapt as attacks change.
- Process - actions aligned with standards such as NIST, PCI and HIPAA, written down so nothing is left out.
- Technology - the right products for detection, prevention and analysis, chosen to fit the organisation's needs and budget.

The roles inside a SOC

Role	Responsibility
SOC Analyst (L1/L2/L3)	Classifies alerts, finds the cause and advises on remediation.
Incident Responder	Performs the initial assessment of a security breach.
Threat Hunter	Proactively hunts threats and vulnerabilities before an attacker can exploit them.
Security Engineer	Builds and maintains the security infrastructure - for example the connection between SIEM and SOAR.
SOC Manager	Handles budgeting, strategy, staff and operations - managerial rather than technical work.

Q1. SOC Types and Roles - complete the lesson.

ANSWER

No answer needed - read the lesson and click Check.

LESSON 3

SOC Analyst and Their Responsibilities

A SOC analyst is the first person to investigate a threat. If the situation demands it, they escalate the incident to a senior so it can be mitigated. Being the first responder makes this one of the most important roles on the team.

The work is varied: there are ever more attack vectors and malware families, so an analyst spends the day investigating different kinds of incidents rather than repeating one task.

A day in the life

An analyst reviews alerts in the SIEM and decides which of them are real threats. To reach a conclusion they use supporting products - EDR, log management and SOAR.

Skills you need to analyse alerts without leaning on the tools

- Operating systems - you must know what normal looks like before you can spot what is abnormal (for example, which Windows services are legitimate).
- Networking - you will deal with malicious IPs and URLs, so you need to confirm whether any device on the network is trying to reach them.
- Malware analysis - enough to determine the command-and-control address of a malicious file and whether anything is talking to it.

Q1. SOC Analyst and Their Responsibilities - complete the lesson.

ANSWER

No answer needed - read the lesson and click Check.

LESSON 4

SIEM and the Analyst Relationship

SIEM stands for Security Information and Event Management - a solution that provides real-time logging of events in an environment so that security threats can be detected.

As a SOC analyst you care about the parts that collect and filter data and raise alerts for suspicious events. Example: if someone on Windows enters 20 wrong passwords in 10 seconds that is suspicious, so a rule is created to alert whenever the threshold is crossed.

Popular SIEM solutions

IBM QRadar, ArcSight ESM, FortiSIEM and Splunk.

Analyst and SIEM

- The SIEM has many features but an analyst mainly tracks alerts; other teams own the configuration and correlation rules.
- In LetsDefend the SIEM is the Monitoring page. New alerts appear in the Main Channel - think of it as a shared channel your teammates can also see.
- Selecting an alert and clicking Take Ownership moves it into the Investigation Channel so the rest of the team can see you are working on it.
- False alerts happen. A rule that raises an alert for URLs containing the word "union" will also fire on a harmless query like <https://www.google.com/search?q=sql+union+usage> - a good analyst reports these so the rule set can be tuned.

Q1. When you close an alert, which channel (tab on the Monitoring page) can you access it from?

ANSWER

Closed Alerts - closed tickets are listed under the Closed Alerts tab.

LESSON 5

Log Management

Log management provides access to all the logs in an environment - web logs, OS logs, firewall, proxy, EDR and more - collected in one place. That saves time and reduces mistakes: without it the same question would have to be sent to a dozen different devices.

Analysts use it mainly to check whether anything is communicating with a suspicious address. If you find that a piece of malware talks to a command-and-control domain, you search that domain in log management to see whether any other device on the network reached it - the original alert may only have flagged one host.

How to solve the next two questions

Open the LetsDefend Log Management page. Click raw_log in the filter, set the filter to raw_log contains, then paste the URL or IP address you are searching for. The matching log rows show the source IP, destination IP, destination port and log type.

Q1. What source IP address entered the URL 'https://github.com/apache/flink/compare'?

ANSWER

172.16.17.54

Q2. What is the type of log that has a destination port number of 52567 and a source IP address of 8.8.8.8?

ANSWER

DNS

LESSON 6

EDR - Endpoint Detection and Response

EDR (also called ETDR - Endpoint Threat Detection and Response) is an endpoint security solution that combines continuous real-time monitoring and collection of endpoint data with automated response and analysis. Common products are CarbonBlack, SentinelOne and FireEye HX.

What you can do with EDR

- The left panel lists accessible endpoint devices; you can search for a host or run an IOC search (file hash, process name, IP) across all hosts.
- The right panel shows device details such as Browser History, Network Connections and the Process List.
- Connect opens the machine itself for a live investigation.
- Containment isolates a compromised device from the internal and external network - it can then only talk to the EDR centre, so you can keep investigating safely.

How to solve the next two questions

Open the LetsDefend Endpoint Security page. For the first question, search the file hash across all hosts and look for nmap.exe - the hostname is shown in the left-hand list. For the second, search for the hostname Roberto first, then open the device and read its CMD history.

Q1. What is the hostname of the device where the "nmap" file with a hash value of "83e0cfc95de1153d405e839e53d408f5" is executed?

ANSWER

EricProd

Q2. A "Ps1.hta" file was executed on a device with the hostname "Roberto". What is the complete CMD command?

ANSWER

C:/Windows/System32/mshta.exe C:/Users/roberto/Desktop/Ps1.hta

LESSON 7

SOAR - Security Orchestration, Automation and Response

SOAR lets the security products in an environment work together and automates the repetitive parts of an analyst's day - for example, looking the source IP of an alert up in VirusTotal before you even open the ticket. Popular products include Splunk Phantom, IBM Resilient, Logsign and Demisto.

Why it matters

- Saves time - workflows automate IP reputation checks, hash queries and sandbox scans.
- Centralisation - one platform gives you the sandbox, log management and third-party tools in a single place.
- Playbooks - step-by-step investigation guides for each scenario, so no step is forgotten and the whole team follows the same process.

In LetsDefend, Case Management plays the role of SOAR: you open a ticket for a case and it comes with an automatically assigned playbook to follow against the related alert.

Q1. SOAR - complete the lesson.

ANSWER

No answer needed - read the lesson and click Check.

LESSON 8

Threat Intelligence Feed

A SOC must know about the latest threats as soon as they appear. A threat intelligence feed is data supplied by a third party - malware hashes, command-and-control domain and IP addresses, and more. Free and popular sources include VirusTotal and Talos Intelligence.

Two traps to avoid

- If a hash or URL is not found in the feeds, that does not make it clean. New malware using antivirus evasion may be undetected - always follow up with static and dynamic analysis.
- IP addresses change hands. An address that once hosted a command-and-control server may later belong to an innocent blog. Past malicious use does not prove present malicious content.

How to solve this question

Open the LetsDefend Threat Intel page and search for the hash. The result card shows which feed the record came from.

Q1. What is the data source of the "e1def6e8ab4b5bcb650037df234e2973" hash on the threat intel page?

ANSWER

AbuseCH

LESSON 9

Common Mistakes Made by SOC Analysts

Everyone makes mistakes; the trick is knowing the four that trip up SOC analysts most often.

Mistake	Why it is wrong and what to do instead
Over-relying on VirusTotal	A "no detections" result is not proof of safety - new malware can bypass antivirus. Treat VirusTotal as a supporting tool and analyse the file yourself.
Hasty sandbox analysis	A 3-4 minute run may show nothing: malware often detects a sandbox, or sleeps for 10-15 minutes before acting. Let the analysis run longer and, where possible, use a real environment.
Inadequate log analysis	Finding one infected host is not the end. Always search the malicious address in log management to see whether other devices are talking to it too.
Overlooking VirusTotal dates	Cached results can be stale - an attacker can point a previously clean URL at malicious content. Start a fresh search rather than trusting the cache.

Q1. Common Mistakes made by SOC Analysts - complete the lesson.

ANSWER

No answer needed - read the lesson and click Check.

LESSON 10

Evaluate Yourself with Quiz

The course ends with a short quiz. All thirteen questions and their answers are listed below.

Q1. Which type of SOC team does not have its own facility and often works remotely in different locations?

ANSWER

Virtual SOC

Q2. I am responsible for connecting security products to each other. What is my job title?

ANSWER

Security Engineer

Q3. What is SOC?

ANSWER

Security Operations Centre

Q4. Which tool is the most important for a SOC analyst? (SIEM / Log Management / EDR / All of this, and much more)

ANSWER

All of this, and much more

Q5. Which type of SOC model consists of internal SOC personnel working with an external Managed Security Service Provider?

ANSWER

Co-Managed SOC

Q6. Which SOC position finds vulnerabilities before the attacker can exploit them in an attack?

ANSWER

Threat Hunter

Q7. What is the goal of a SIEM?

ANSWER

To provide real-time logging of events in an environment (so threats can be detected)

Q8. Which LetsDefend page is the SIEM?

ANSWER

Monitoring

Q9. What is an EDR?**ANSWER**

Software that monitors the terminals (computers, servers, tablets, phones) and not the information system network

Q10. What can be used to conduct an analysis with SOAR?**ANSWER**

Playbook

Q11. Which information do you not have in the Threat Intelligence Feed?**ANSWER**

A sample of the infected file

Q12. Which is a common mistake for SOC analysts?**ANSWER**

Insufficient log analysis

Q13. What are the different steps of the NIST incident lifecycle?**ANSWER**

Preparation, Detection/Analysis, Containment/Eradication and Recovery, Post-Event Activity

Well done

Answering all thirteen correctly completes the quiz, finishes the course and unlocks the "SOC Member" badge.

Answer key at a glance

Every question in the course, in one place. Try to answer each one yourself first, then use this table to check.

Lesson	Question	Answer
4	Channel an alert is accessed from once closed	Closed Alerts
5	Source IP that entered the apache/flink compare URL	172.16.17.54
5	Log type with destination port 52567 from 8.8.8.8	DNS
6	Hostname where the nmap hash is executed	EricProd
6	Full CMD command for Ps1.hta on host Roberto	C:/Windows/System32/mshta.exe C:/Users/roberto/Desktop/Ps1.hta
8	Data source of the threat intel hash	AbuseCH
Quiz	Type of SOC that works remotely	Virtual SOC
Quiz	Role that connects security products	Security Engineer
Quiz	What SOC stands for	Security Operations Centre
Quiz	Most important tool for a SOC analyst	All of this, and much more
Quiz	Internal SOC staff plus an external MSSP	Co-Managed SOC
Quiz	Role that finds vulnerabilities first	Threat Hunter
Quiz	Goal of a SIEM	Real-time logging of events in an environment
Quiz	LetsDefend page that is the SIEM	Monitoring
Quiz	What an EDR is	Monitors terminals, not the network
Quiz	What is used to analyse with SOAR	Playbook
Quiz	Not available in the Threat Intelligence Feed	A sample of the infected file
Quiz	Common mistake for SOC analysts	Insufficient log analysis
Quiz	NIST incident lifecycle steps	Preparation, Detection/Analysis, Containment/Eradication and Recovery, Post-Event Activity

Next steps

You now know what a SOC is, how it is organised and which tools a SOC analyst depends on. Continue with the Cyber Kill Chain to learn how attacks are staged, then MITRE ATT&CK and SIEM 101 to put the alerting side into practice.